AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M238
Data Visualization

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed:** (1.5) Hours

Instructions:

- Total Marks: **30**
- This exam consists of **TWO** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1

TRUE or FALSE (5 marks)

MARK the following statements are TRUE or FALSE

1. Data visualization helps people see and understand patterns in data.
2. Data visualization and data storytelling are the same thing within the broader framework of visual analytics.
3. Bars in a bar chart can only be oriented on the vertical axis.
4. In bar charts, it is best practice to sort your data in either ascending or descending order, depending on the use case.
5. Annotations in a visualization are used to highlight specific marks, points, or areas.

PART 2

ALL QUESTIONS ARE REQUIRED (25 marks)

1. What is data visualization? (2 Marks)
2. What type of data is the bar chart best suited for, and what is its primary purpose? (2 Marks)
3. Define what a line chart (or time series chart) is and explain how it visualizes data. (2 Marks)
4. Why is emotional appeal alone not enough to create a meaningful visual data story? (2 Marks)
5. How are the four levels of measurement classified in terms of discrete and continuous data? (4 Marks)
6. What is the main purpose of filters in Tableau? (2 Marks)
7. What are scatter plots best used for in data visualization? (2 Marks)
8. How does Tableau determine which geographic field to plot when multiple geographic fields are placed on the Detail list? (2 Marks)
9. List the Pros and Cons of Live versus Extract connection options. (7 Marks)

End of questions

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